KAIZEN MATHS

Lesson starter

Upper and Lower Bounds

Exam-Style Bounds

Name _____

S1

Find the maximum percentage error for each rounded value.

A measurement is 11 g to the nearest whole number. Find the maximum percentage error.

S2

State the rounded value and the accuracy represented by each interval.

The error interval for x is $25 \text{ s} \leq x < 35 \text{ s}$. State the rounded value and the accuracy.

S3

Find the maximum percentage error for each rounded value.

A measurement is 45.79 g to the nearest 0.01. Find the maximum percentage error.

S4

State the rounded value and the accuracy represented by each interval.

The error interval for x is $64.995 \text{ cm} \leq x < 65.005 \text{ cm}$. State the rounded value and the accuracy.

Show your working

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NUM-E608-L4 · S

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KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Exam-Style Bounds

Name

EA1

Find the maximum percentage error for each rounded value.

A measurement is 27.4 cm to the nearest 0.1. Find the maximum percentage error.

EA2

State the rounded value and the accuracy represented by each interval.

The error interval for x is 54.905 cm ≤ x < 54.915 cm . State the rounded value and the accuracy.

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KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Exam-Style Bounds

Name

EB1

Find the maximum percentage error for each rounded value.

A measurement is 44.6 cm to the nearest 0.1. Find the maximum percentage error.

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KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Exam-Style Bounds

Name

EA1

Find the maximum percentage error for each rounded value.

A measurement is 27.4 cm to the nearest 0.1. Find the maximum percentage error.

EA2

State the rounded value and the accuracy represented by each interval.

The error interval for x is 54.905 cm ≤ x < 54.915 cm . State the rounded value and the accuracy.

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KAIZEN MATHS

Exit ticket

Upper and Lower Bounds

Exam-Style Bounds

Name

EB1

Find the maximum percentage error for each rounded value.

A measurement is 44.6 cm to the nearest 0.1. Find the maximum percentage error.

NUM-E608-L4 · EB

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Teacher answers & working

Upper and Lower Bounds · Exam-Style Bounds

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S1

Find the maximum percentage error for each rounded value.

A measurement is 11 g to the nearest whole number. Find the maximum percentage error.

Answer:

4.545%

Working:

Maximum absolute error = half the rounding unit.

Maximum absolute error = 0.5 g.

Percentage error = $\frac{maximumabsoluteerror}{measuredvalue}$ × 100%.

Percentage error = $\frac{0.5}{11}$ × 100% = 4.545%.

S2

State the rounded value and the accuracy represented by each interval.

The error interval for x is 25 s ≤ x < 35 s . State the rounded value and the accuracy.

Answer:

30 s, to the nearest 10

Working:

The rounded value is the midpoint of the interval.

Midpoint = $\frac{25+35}{2}$ = 30.

The interval width is 10 s.

So the value was rounded to the nearest 10.

Rounded value = 30 s.

S3

Find the maximum percentage error for each rounded value.

A measurement is 45.79 g to the nearest 0.01. Find the maximum percentage error.

Answer:

0.011%

Working:

Maximum absolute error = half the rounding unit.

Maximum absolute error = 0.005 g.

Percentage error = $\frac{maximumabsoluteerror}{measuredvalue}$ × 100%.

Percentage error = $\frac{0.005}{45.79}$ × 100% = 0.011%.

Teacher answers & working

Upper and Lower Bounds · Exam-Style Bounds

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S4

State the rounded value and the accuracy represented by each interval.

The error interval for x is 64.995 cm ≤ x < 65.005 cm . State the rounded value and the accuracy.

Answer:

65 cm, to the nearest 0.01

Working:

The rounded value is the midpoint of the interval.

Midpoint = $\frac{64.995+65.005}{2}$ = 65.

The interval width is 0.01 cm.

So the value was rounded to the nearest 0.01.

Rounded value = 65 cm.

EA1

Find the maximum percentage error for each rounded value.

A measurement is 27.4 cm to the nearest 0.1. Find the maximum percentage error.

Answer:

0.182%

Working:

Maximum absolute error = half the rounding unit.

Maximum absolute error = 0.05 cm.

Percentage error = $\frac{maximumabsoluteerror}{measuredvalue}$ × 100%.

Percentage error = $\frac{0.05}{27.4}$ × 100% = 0.182%.

EA2

State the rounded value and the accuracy represented by each interval.

The error interval for x is 54.905 cm ≤ x < 54.915 cm . State the rounded value and the accuracy.

Answer:

54.91 cm, to the nearest 0.01

Working:

The rounded value is the midpoint of the interval.

Midpoint = $\frac{54.905+54.915}{2}$ = 54.91.

The interval width is 0.01 cm.

So the value was rounded to the nearest 0.01.

Rounded value = 54.91 cm.

Teacher answers & working

Upper and Lower Bounds · Exam-Style Bounds

NUM-E608-L4 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

EB1

Find the maximum percentage error for each rounded value.

A measurement is 44.6 cm to the nearest 0.1. Find the maximum percentage error.

Answer:

0.112%

Working:

Maximum absolute error = half the rounding unit.

Maximum absolute error = 0.05 cm.

Percentage error = $\frac{maximumabsoluteerror}{measuredvalue}$ × 100%.

Percentage error = $\frac{0.05}{44.6}$ × 100% = 0.112%.